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Internet Review

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Data Link Interconnection

- Internet is the interconnection of different Data Link Layer (DLL) technologies.
- IP is the protocol that allows this interconnection.
- DLL on their own do not provide universal interconnection:
  - Different frame formats, address formats, etc.
  - Are not designed to scale to millions of interconnected devices. E.g. MAC address organization and MAC learning in Ethernet.

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Addresses

- In general, L2 or MAC addresses are not placed in the network according to any specific policy.
- The only constrain is that hardware addresses must be unique inside the link or data link network.
- In Ethernet and WLAN the MAC address has 6 bytes (e.g. 08:0B:F0:AF:DC:09).
- IP addresses are independent of any data link technology.
- They uniquely and universally identify a network device in the Internet.
- An IP address is logically assigned (configured) to a NIC.
- IP addresses are assigned following some “grouping” policy or “**network addressing plan**” to be able to aggregate them in the routing process.

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Addressing Plan

- IP divides the address in two parts: **NetID** and **HostID**.
- To be scalable, IP provides an addressing plan or a way of assigning IP addresses.
- The basic rule for assigning IP addresses is that:

IP Addressing Plan

All the devices of a data link must use IP addresses with the same NetID.

- In this way, we can aggregate for IP routing all the devices that are “near”, i.e. in the same L2.
- To make the division of the IP address we use network masks, which are a local parameter of each IP device.

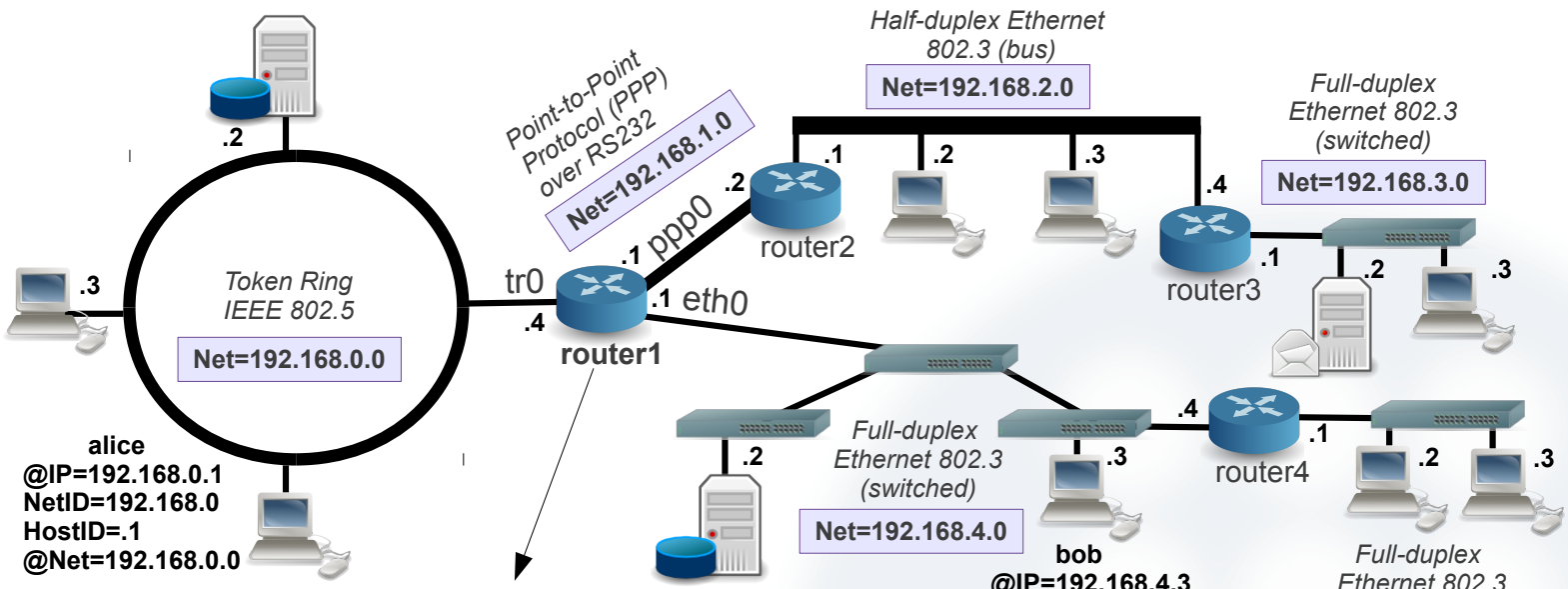
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IP Routing I

- IP uses the **addressing plan** mentioned to know if a device is the local data link.
- For devices that are not in a local link, IP uses a **hop by hop** routing mechanism:
  - Each IP device only must take care of knowing the appropriate parameters to send the datagram to the next node (next hop) in the path to the destination.
- IP is **scalable** compared with Ethernet because:
  - **Routing tables group addresses:**
    - The amount of information that a router needs to keep is proportional to the number of networks in the Internet, not the number of devices.
  - **Routers filter L2 traffic:**
    - Routers only forward (route) IP packets between data link networks.
    - Frames which do not contain IP are not forwarded.

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IP Routing II



Dst Net	Next Hop
192.168.0.0/24	tr0, 0.0.0.0
192.168.1.0/24	ppp0, 0.0.0.0
192.168.2.0/24	ppp0,0.0.0.0
192.168.3.0/24	ppp0, 0.0.0.0
192.168.4.0/24	eth0, 0.0.0.0
192.168.5.0/24	eth0, 192.168.4.4

Routing table of router1

IP	MAC
192.168.0.1	11:22:33:44:55:00
192.168.0.2	11:22:33:44:55:11
192.168.0.3	11:22:33:44:55:22
192.168.4.2	11:22:33:44:55:33
192.168.4.3	11:22:33:44:55:44
192.168.4.4	11:22:33:44:55:55

IP/MAC Table of router1 (ARP cache)

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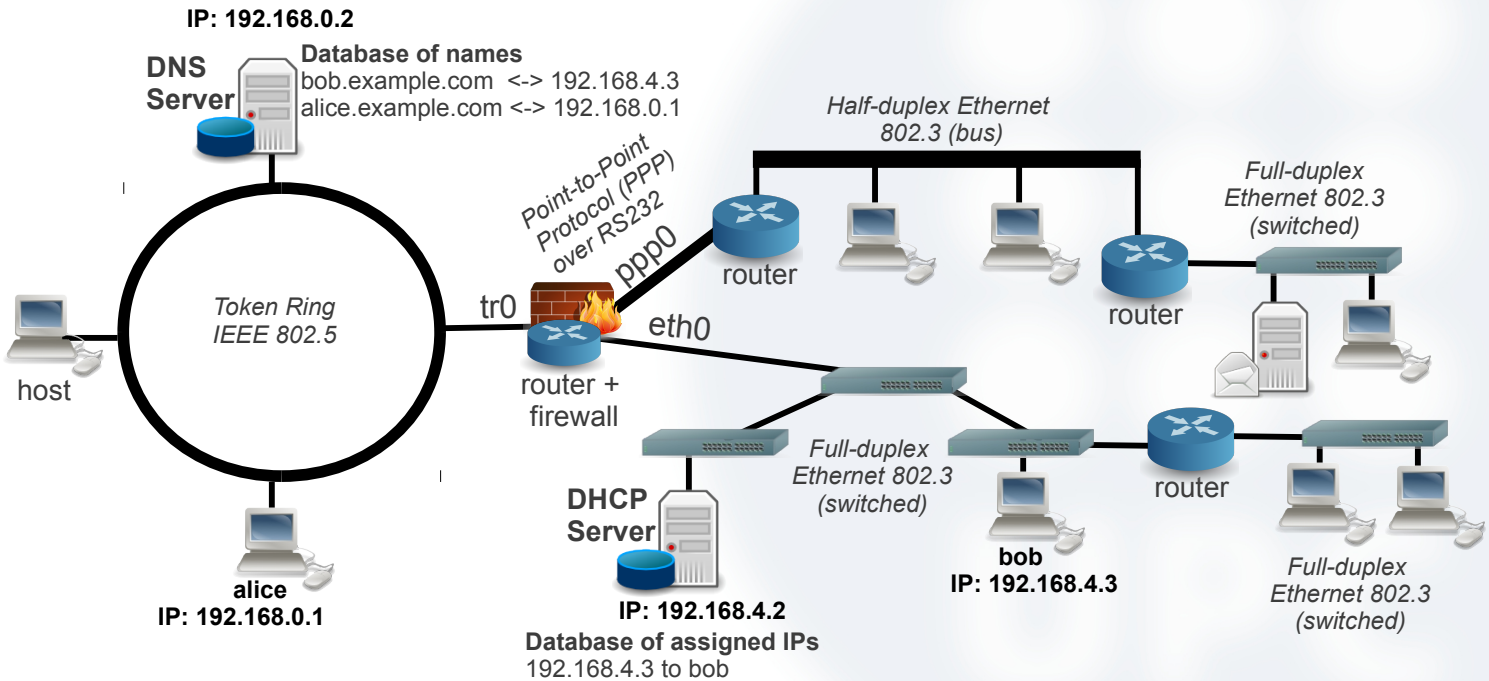
Names & Dynamic IP

- **Dynamic IPs.**
  - Nowadays it is not usual to manually configure your IP addresses.
  - Instead, a server and a protocol called **DHCP (Dynamic Host Configuration Protocol)** are used.
- **Names.**
  - Today, it is neither common to use IP addresses to refer to IP destinations.
  - Instead of IP addresses, which are hard to remember, users utilize names like in your phone “address book”.
  - The “address book” for the Internet names is placed in servers and a protocol is used to access them.
  - This system is called **DNS (Domain Name System)**.

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Firewalls

- Routers only forward IP packets (not frames without IP).
- A router with “firewall“ capabilities can filter (do not forward) other traffic.
- Firewalls take filtering decisions based on protocols and protocol parameters.



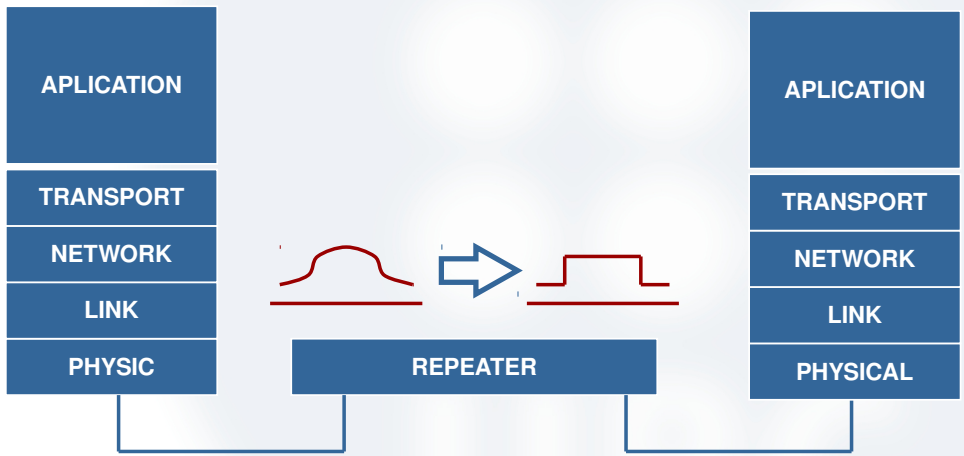
- E.g. filter UDP port 53 (DNS service).

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Repeater

- Operates at physical layer (L1) restoring the signal.
- It can perform some signal modifications (e.g. amplification).
- It is not able to read or store bits of information.
- As it works in physical layer, only same kind data links can be interconnected with repeaters.

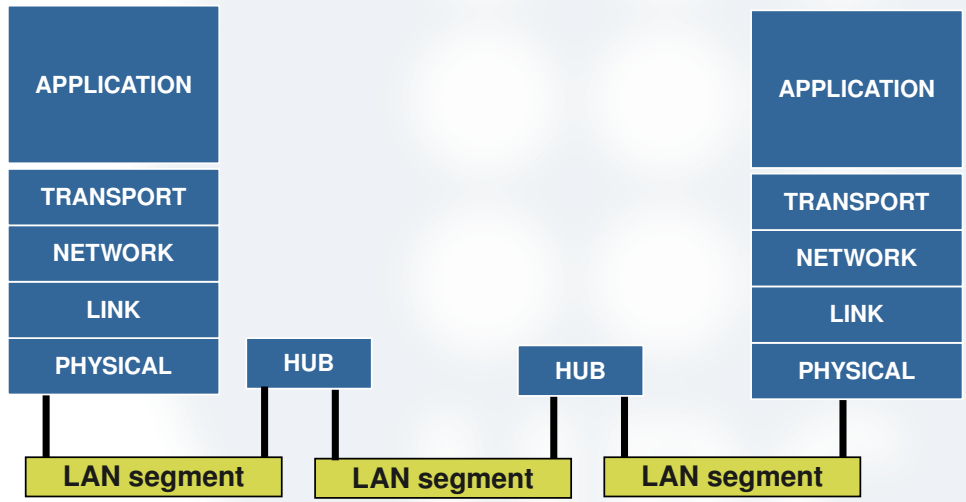
- For each type of network there are rules which define:
  - Maximum distance between repeaters.
  - Maximum number of repeaters in the data link.



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Hub

- A Hub is a multiport repeater.
- It creates a bus for multiple point to point connections (typically copper pairs).
- It receives signal from all connected stations.
- It forwards everything it receives to all hosts connected to the hub.
- All hosts share the same bandwidth.
- Hubs can be interconnected.



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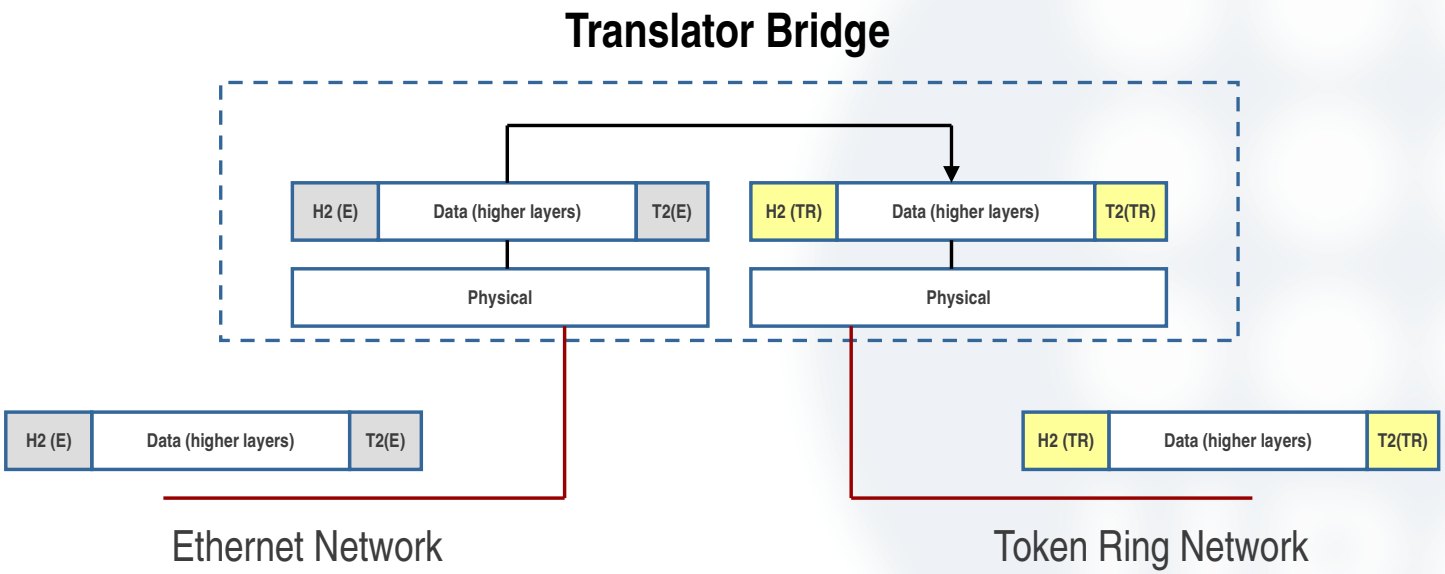
Bridge I

- A Bridge Has two ports and operates at layer 2.
- Bridges are able to analyze MAC Addresses.
- Frames are forwarded based on the MAC address.
- They can be interconnected.
- There are 2 types of bridges: transparent and translator.
- **Transparent Bridge**:
  - Connects multiple network segments at the data link layer (layer 2).
  - They do not alter the frames that pass through them.
  - Source and destination ignore that there is a bridge in the middle.
  - Uses MAC learning.

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Bridge II

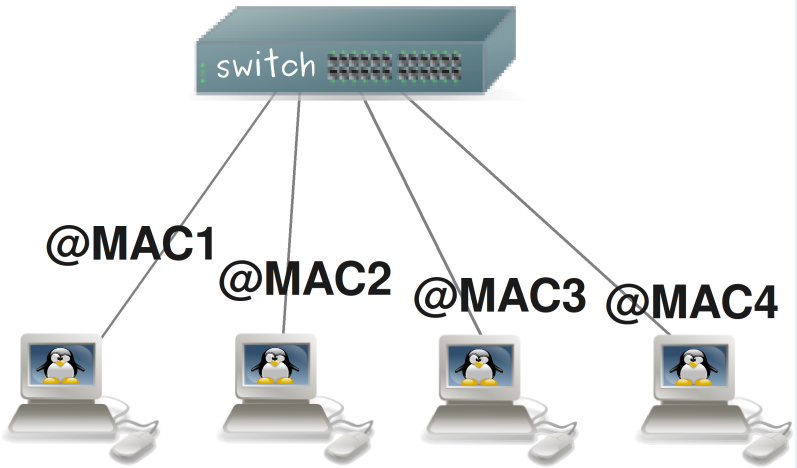
- **Translator Bridge**:
  - These bridges also analyze the frame format.
  - They can interconnect networks which use different formats.
  - To do so, they translate appropriately these formats.



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Switch

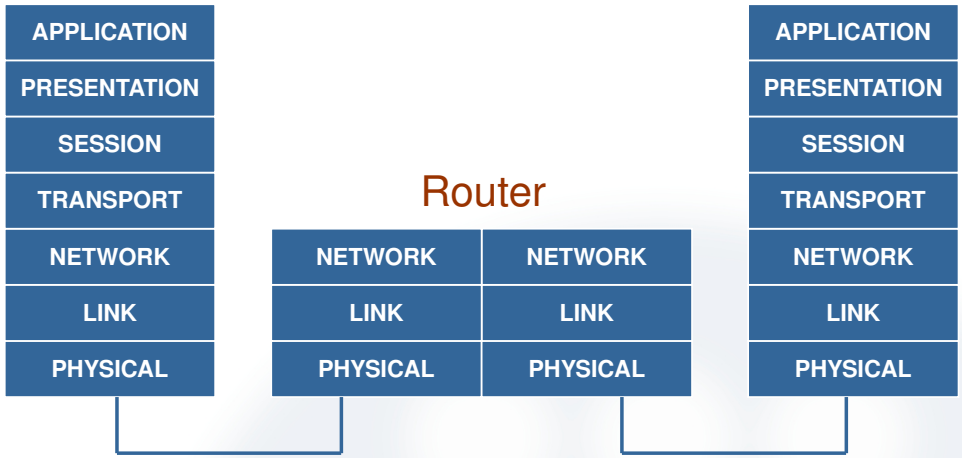
- A switch can be defined as a multi-port bridge.
- They avoid collisions because the remove the idea of shared media.
- Switches use MAC learning.
- They can be interconnected but if the topology has cycles we need the spanning tree protocol.

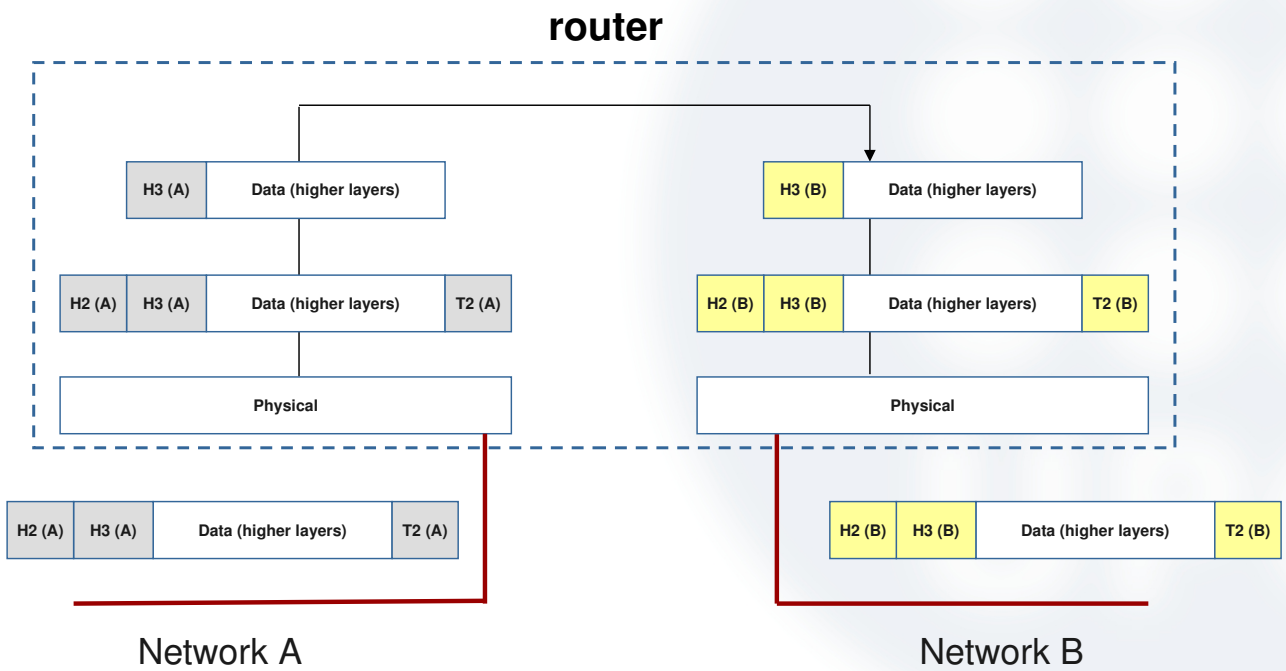


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Router

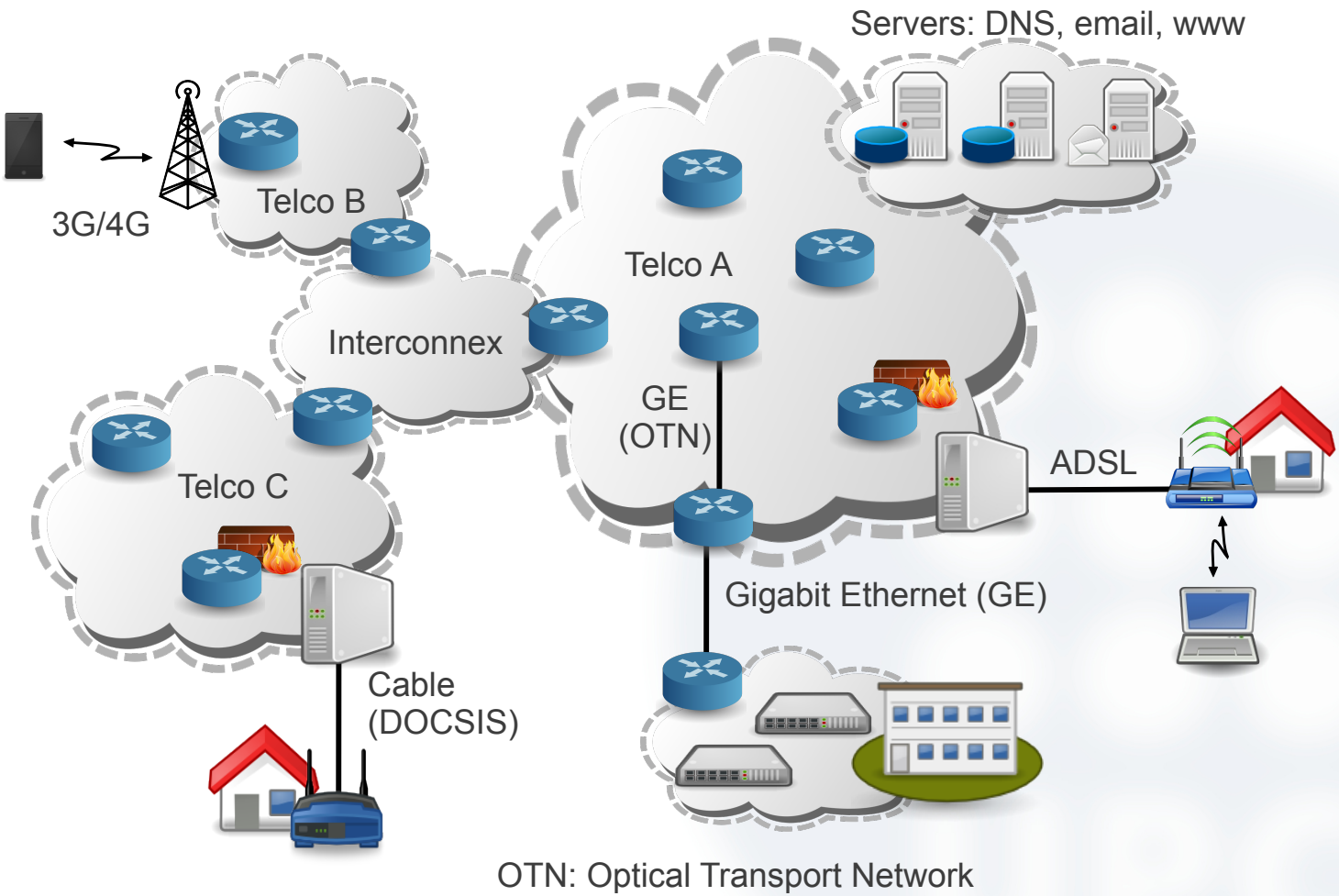
- Forwards of IP packets.
- Analyzes the IP packet and uses a routing table to do so.





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Internet is not so Simple...



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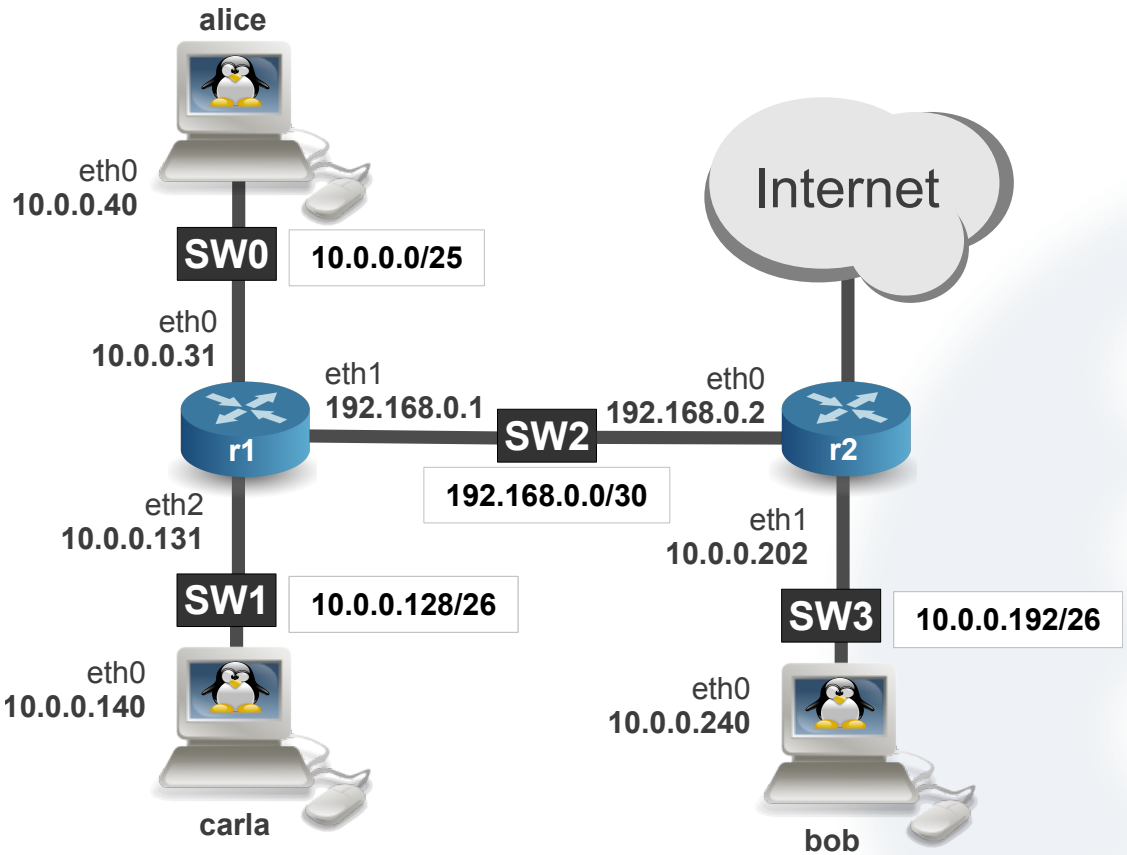
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Why Virtual Networks? I



- If you want to implement this topology, you need a certain investment for creating the network with real equipment: switches, cables, routers, hosts, servers etc.

- You also need to implement the physical configuration of the devices: connect cables, etc.

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Why Virtual Networks? II

- With the technologies for building virtual networks, we just need:
  - A **single computer** (physical host).
  - A **VNUML file to define the network topology (scenario)**.
- VNUML comes from Virtual Network User Mode Linux (http://www.dit.upm.es/vnuml).
- With **simtools** (a wrapper of VNUML) we can build any network topology with **hubs**, **switches** and **UMLs (Linux machines)**.
- **Linux boxes are very flexible:** they can be used as routers, hosts and as nearly any server (DNS, DHCP, WWW, etc.).

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Why Virtual Networks? III

- In a virtual network everything is implemented (real) except that hub/switches are software processes.
- Thus, bits are not transmitted through physical media but they are just moved inside a computer.
- The **phyhost** is connected to “virtual switches“ of the virtual network using special NICs called `taps` that are created in the **phyhost** when the scenario is started.
- We will use pseudo-terminal(s) to access to each VM.

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Install & Update Simtools

- The installation has been tested using the **32-bit version** of Ubuntu 12.04.

Important: `simctl` does not work with the 64-bit version.

- To install simtools, once you have a Linux box (**phyhost**) running, type the following commands:

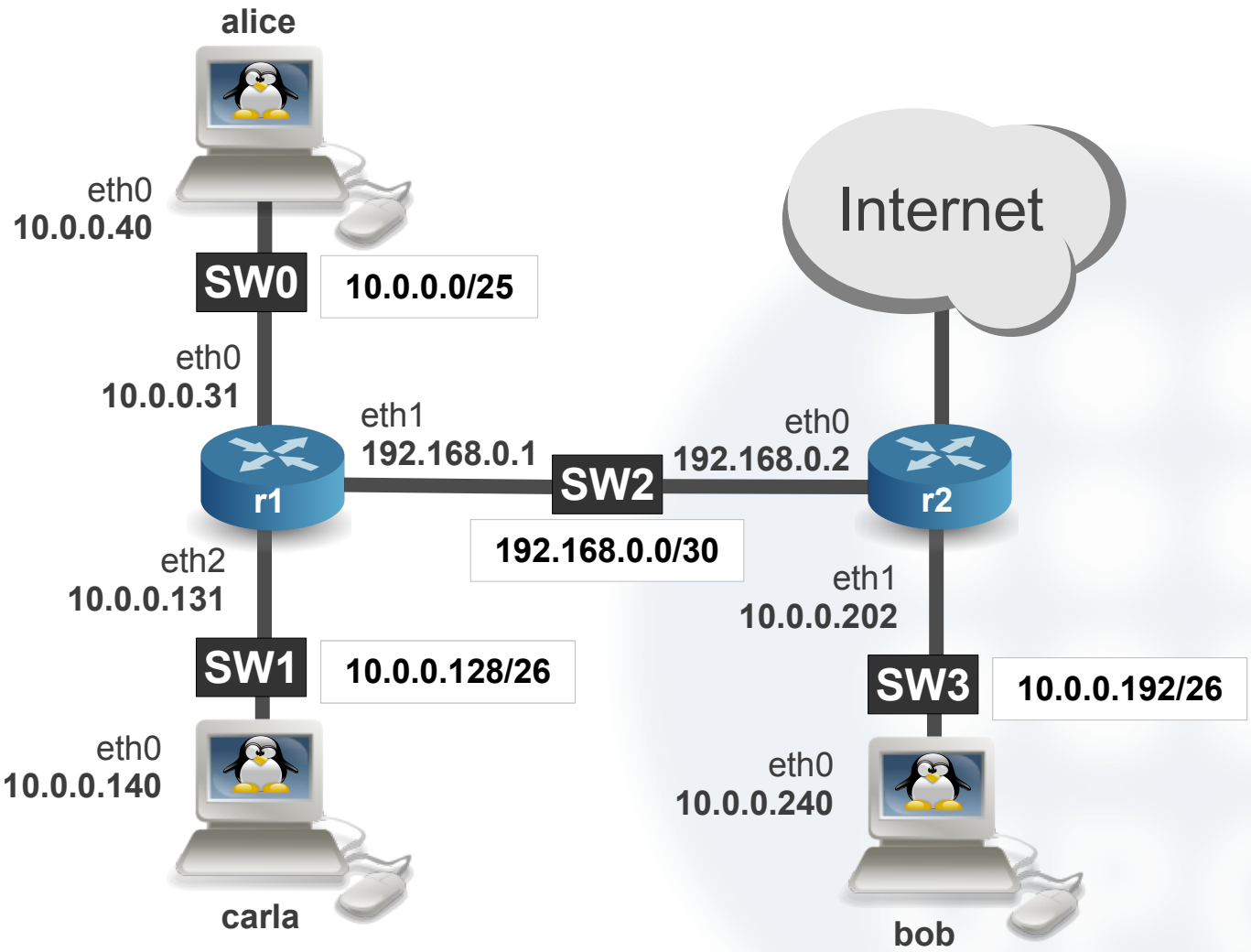
```
phyhost$ echo "deb http://sertel.upc.es/~vanetes/debian i386/" |
sudo tee /etc/apt/sources.list.d/simtools.list
phyhost$ sudo apt-get update
phyhost$ sudo apt-get install metasimtools -y --force-yes
```

- You can repeat these steps if the software is not installed correctly at the first time.
- To update the software type:

```
phyhost$ simctl forcestop
phyhost$ sudo apt-get update
phyhost$ sudo apt-get install metasimtools* -y --force-yes
```

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Example Topology



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Scenario's VNUML File

```
<?xml version="1.0" encoding="UTF−8"?>
<!DOCTYPE vnuml SYSTEM "/usr/share/xml/vnuml/vnuml.dtd">
<vnuml>
  <global>
    <version>1.8</version>
    <simulation_name>routing−abc</simulation_name>
    <automac/>
    <vm_mgmt type="none" />
    <vm_defaults exec_mode="mconsole">
      <filesystem type="cow">/usr/share/vnuml/filesystems/root_fs_tutorial</filesystem>
      <kernel>/usr/share/vnuml/kernels/linux</kernel>
      <console id="0">pts</console>
    </vm_defaults>
  </global>
  <net name="Net0" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net0.ctl" />
  <net name="Net1" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net1.ctl" />
  <net name="Net2" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net2.ctl" />
  <net name="Net3" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net3.ctl" />
  <vm name="alice">
    <if id="0" net="Net0"></if>
  </vm>
  <vm name="r1">
    <console id="1">pts</console>
    <if id="0" net="Net0"></if>
    <if id="1" net="Net2"></if>
    <if id="2" net="Net1"></if>
    <forwarding type="ip" />
  </vm>
  ...
</vnuml>
```

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Locating Scenarios

- The scenario files must have the extension .vnuml
- By default, `simctl` searches vnuml files in /usr/share/vnuml/scenarios.
- You can place these files in any directory using the variable DIRPRACT:

```
phyhost$ export DIRPRACT=.
phyhost$ simctl
simctl ( routing-abc ) (OPTIONS)
OPTIONS
    start                          Starts scenario
    stop                           Stops scenario
    status                         State of simulation
                                   (running | stopped)
    vms                            Shows vms from simulation
    labels [vm]                    Shows sequence labels
                                   for ALL vms or for vm
    exec label [vm]                Exec label in the vms
                                   where label is defined or
                                   exec the label only in the vm
    netinfo                        Provides info about
                                   network connections
    get [-t term_type] vm [pts]    Gets a terminal for vm
                                   term_type selects terminal
                                   (xterm | gnome | kde | local)
                                   pts is an integer
                                   to select the console
```

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Start a Scenario

- To start a particular scenario, you must execute `simctl` in the host with the name of the selected scenario and use the start option. Example:

```
phyhost$ simctl routing-abc start
start
 name="tap0" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net0.ctl"
 name="tap1" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net1.ctl"
 name="tap2" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net2.ctl"
 name="tap3" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net3.ctl"

Set 'tap0' persistent and owned by uid 1000
Set 'tap1' persistent and owned by uid 1000
Set 'tap2' persistent and owned by uid 1000
Set 'tap3' persistent and owned by uid 1000
host> /usr/bin/touch /home/user/.vnuml/LOCK
....

Total time elapsed: 77 seconds
phyhost$
```

- Be patient because it might take some time to complete the starting process[1].
- Finally, the command ends indicating the time taken to start the scenario and we get the console prompt again.

[1] Starting an scenario might take up to several minutes.

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Access to VMs I

- You can use the `vms` option to view the Virtual Machines (VMs) of an scenario:

```
phyhost$ simctl routing-abc vms
Virtual machines from routing-abc:
num vms        enabled tty's Id
  1 alice      0
  2 r1         0 1
  3 r2         0
  4 bob        0
  5 carla      0
```

- In this scenario, we can see that all the VMs have a single console (console 0) except **r1**, which has two consoles enabled (consoles 0 and 1).

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Access to VMs II

- You can get a console of a VM with the `get` option:

```
phyhost$ simctl routing-abc get
    alice    Running    --------
    r1       Running    --------
    r2       Running    --------
    bob      Running    --------
    carla    Running    --------
phyhost$ simctl routing-abc get alice
phyhost$ simctl routing-abc get r1
phyhost$ simctl routing-abc get r1 1
phyhost$ simctl routing-abc get r2 0
```

- You will observe that a terminal appears in your system each time you execute the `get` command for a VM.
- To login as root in a VM type:
  - **ENTER**.
  - Use the **root** user with **xxxx** as password.

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Stop a Scenario

- When you wish to stop the simulation, you can type the following:

```
phyhost$ simctl routing-abc stop
host> /usr/bin/touch /home/user/.vnuml/LOCK
host> /bin/rm -f ~/.vnuml/simulations/routing-abc/vms/alice/status
...
---------- Waiting until UML extinction ----------
waiting on processes 6719 7482 8583 9351 10288...
15 seconds elapsed...
host> /bin/rm -f /home/user/.vnuml/networks/Net0.ctl.cter
....
Total time elapsed: 15 seconds
stop
 name="tap0" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net0.ctl"
 name="tap1" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net1.ctl"
 name="tap2" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net2.ctl"
 name="tap3" mode="uml_switch" hub="yes" sock="/var/run/vnuml/Net3.ctl"
Set 'tap0' nonpersistent
Set 'tap1' nonpersistent
Set 'tap2' nonpersistent
Set 'tap3' nonpersistent
phyhost$
```

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Troubleshooting I

- If there is a problem starting a simulation you should type the following commands to clear the system and start it again:

```
phyhost$ simctl simulation_name stop
phyhost$ simctl forcestop
phyhost$ simctl simulation_name start
```

- You should **never run two different simulations at the same time!!!**
- Stop a simulation to run another one.
- If by mistake you start two simulations do:

```
phyhost$ simctl simulation_name1 stop
phyhost$ simctl simulation_name2 stop
phyhost$ simctl forcestop
```

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Troubleshooting II

- You should **never use the superuser ”root“ to execute `simctl`!!!**
- If by mistake you start a simulation with the root user, you must clear the system and start it again using your user:

```
phyhost$ sudo -s
phyhost# simctl simulation_name stop
phyhost# simctl forcestop
phyhost# exit
phyhost$ simctl simulation_name start
```

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Terminal Customization

- Copy the file `/usr/local/etc/simrc` to `~/.simrc`.
- With this file you can define to use `gnome-terminal` instead of `xterm` setting the TERM_TYPE variable:

```
# simrc: tunning of environment variables for simctl
# Definition of scenario files directory
DIRPRACT=/usr/share/vnuml/scenarios
# Definition of VNUML working directory
# (default to $HOME/.vnuml)
# VNUMLWORKDIR=/tmp/$USER
# Definiton of the user which launch the
# switch software
TAPUSER=$USER
# Change the default terminal type (xterm)
# values: (gnome | kde | local)
TERM_TYPE=gnome
# XTERM_FG=white
# XTERM_BG=black
```

- Then, with a `gnome-terminal` you can create a profile called **vnuml** to define colors, font sizes, etc.

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Problems and Workarounds

- **Problem**. The UML Kernel is not aware of terminal size.
  - The consequence of this is that when you resize the terminal, the size of the terminal is not refreshed.
  - **Workaround**. Once in a virtual machine terminal and when the terminal size is modified, keypress Ctrl+a f.
- **Problem**. There is a lack of terminal scrolling.
  - We use `screen`, which has a scrollback history buffer for each virtual terminal of 100 lines.
  - **Workaround**. To enter screen into scrolback mode press Ctrl+a ESC.
    To exit scrollback mode press ESC.

Outline

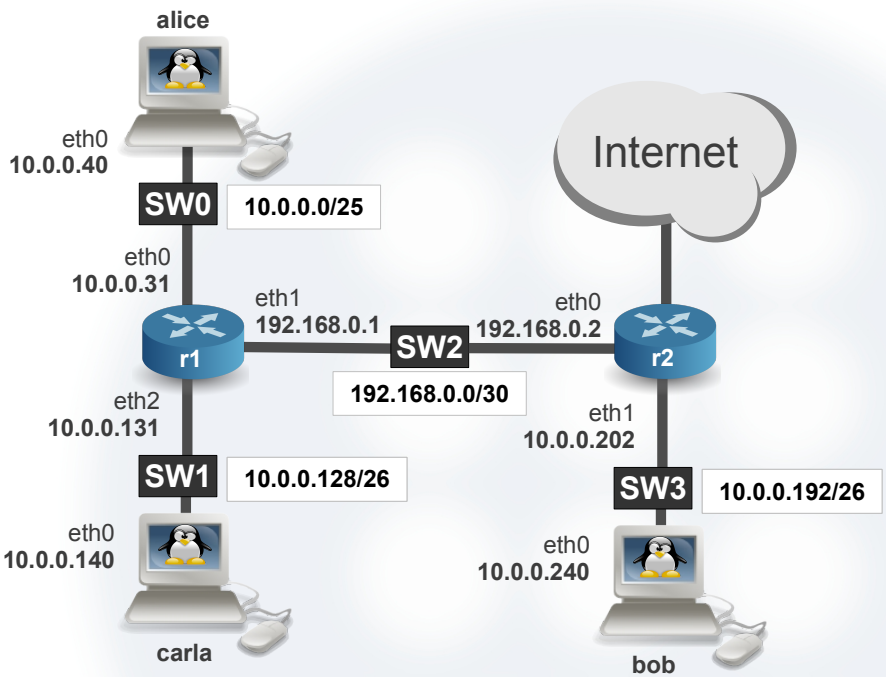
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Example Topology

- We are going to configure the interfaces and routes of this IP network.
- The configuration commands shown are valid for any Unix-like system.
- Our first goal is to enable pings between **alice** and **bob**.



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Interface Configuration I

- Let's start configuring **alice**.
- If you want to remove any @IP previously configured in an interface:

```
alice:~# ifconfig 0.0.0.0 eth0
alice:~# ifconfig
eth0      Link encap:Ethernet  HWaddr fe:fd:00:00:01:00
          inet6 addr: fe80::fcfd:ff:fe00:100/64 Scope:Link
          UP BROADCAST RUNNING PROMISC MULTICAST  MTU:1500  Metric:1
          RX packets:27 errors:0 dropped:0 overruns:0 frame:0
          TX packets:6 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3582 (3.4 KiB)  TX bytes:468 (468.0 B)
          Interrupt:5

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:10 errors:0 dropped:0 overruns:0 frame:0
          TX packets:10 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:660 (660.0 B)  TX bytes:660 (660.0 B)
```

- You can check that currently, the routing table is empty:

```
alice:~# route -n
Kernel IP routing table
Destination     Gateway     Genmask   Use Iface
```

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Interface Configuration II

- Let's configure an IP address and a mask for **alice**'s `eth0`:

```
alice:~# ifconfig eth0 10.0.0.40 netmask 255.255.255.128
alice:~# ifconfig eth0
eth0      Link encap:Ethernet  HWaddr fe:fd:00:00:01:00
          inet addr:10.0.0.40  Bcast:10.0.0.127  Mask:255.255.255.128
          inet6 addr: fe80::fcfd:ff:fe00:100/64 Scope:Link
          UP BROADCAST RUNNING PROMISC MULTICAST  MTU:1500  Metric:1
          RX packets:28 errors:0 dropped:0 overruns:0 frame:0
          TX packets:6 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3669 (3.5 KiB)  TX bytes:468 (468.0 B)
          Interrupt:5
```

- If we check **alice**'s routing table:

```
alice:~# route -n
Kernel IP routing table
Destination  Gateway  Genmask         Flags Metric Ref  Use Iface
10.0.0.0     0.0.0.0  255.255.255.128 U     0      0      0 eth0
```

- As you can observe, the `ifconfig` command not only configures the address and mask but also configures the routing table with the corresponding entry for direct delivery.
- In this case, the entry for the network 10.0.0.0/25.

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Configuration of Routes I

- In the scenario, the **phyhost** has interfaces `tapX` connected to switches **SWX** in the scenario.
- Let's ping **bob**:

```
alice:~# ping -c 1 10.0.0.240
connect: Network is unreachable
```

- The previous error means that there is not any valid entry in **alice**'s routing table for the destination IP address.
- So, we will see nothing if we capture on `tap0`.
- Thus, we need to define an indirect route in **alice**.
- We can use the `route` command which is present at any Unix-like system to add, remove or view routes.
- The most commonly used syntax of route is the following:

```
# route (add|del) -net @NET netmask MASK [gw @IP dev IF]
```

- We can also use the CIDR notation `@NET/X`.
- We can view the current routing table with the `-n` option, which means view the routing table numerically (without name resolution).

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Configuration of Routes II

- We can add routing entries for hosts with the `-host` option.
- A host route is the most specific entry that can be used since the destination must match completely the 32 bits of the routing entry.
- Finally, we can define a default route with:

```
route add default gw @IP
```

- Let's configure an indirect route to **bob**'s network:

```
alice:~# route add -net 10.0.0.192/26 gw 10.0.0.31
alice:~# route -n
Kernel IP routing table
Destination     Gateway     Genmask         Flags Metric Ref    Use Iface
10.0.0.192      10.0.0.31   255.255.255.192 UG    0      0        0 eth0
10.0.0.0        0.0.0.0     255.255.255.128 U     0      0        0 eth0
alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
From 10.0.0.40 icmp_seq=1 Destination Host Unreachable

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms
```

- Now the problem is that the `eth0` NIC of **r1** is not properly configured.

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Conf. of **alice** (Cont) I

- A golden rule for configuring routes is that **you cannot configure an indirect route using a router to which you are not directly connected**:

```
alice:~# route -n
Kernel IP routing table
Destination     Gateway     Genmask         Flags Metric Ref    Use Iface
10.0.0.192      10.0.0.31   255.255.255.192 UG    0      0        0 eth0
10.0.0.0        0.0.0.0     255.255.255.128 U     0      0        0 eth0
alice:~# route add -net 10.0.0.128/26 gw 10.0.0.131
SIOCADDRT: Network is unreachable
alice:~# route add -net 10.0.0.128/26 gw 10.0.0.202
SIOCADDRT: Network is unreachable
```

- Let's set up the indirect route for **carla**'s network correctly:

```
alice:~# route add -net 10.0.0.128/26 gw 10.0.0.31
alice:~# route -n
Kernel IP routing table
Destination     Gateway     Genmask         Flags Metric Ref    Use Iface
10.0.0.128      10.0.0.31   255.255.255.192 UG    0      0        0 eth0
10.0.0.192      10.0.0.31   255.255.255.192 UG    0      0        0 eth0
10.0.0.0        0.0.0.0     255.255.255.128 U     0      0        0 eth0
```

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Conf. of **alice** (Cont) II

- We can also define specific routes for particular hosts:

```
alice:~# route add -host 10.0.0.140 gw 10.0.0.31
alice:~# route -n
Kernel IP routing table
Destination     Gateway     Genmask         Flags Metric Ref    Use Iface
10.0.0.140      10.0.0.31   255.255.255.255 UGH   0      0        0 eth0
10.0.0.128      10.0.0.31   255.255.255.192 UG    0      0        0 eth0
10.0.0.192      10.0.0.31   255.255.255.192 UG    0      0        0 eth0
10.0.0.0        0.0.0.0     255.255.255.128 U     0      0        0 eth0
```

- Notice that **r1** is the only node that can be used by **alice** to forward her indirect traffic.
- Thus, we can delete the previous indirect routes and add a default route:

```
alice:~# route del -host 10.0.0.140
alice:~# route del -net 10.0.0.128/26
alice:~# route del -net 10.0.0.192/26
alice:~# route add default gw 10.0.0.31
alice:~# route -n
Kernel IP routing table
Destination     Gateway     Genmask         Flags Metric Ref    Use Iface
10.0.0.0        0.0.0.0     255.255.255.128 U     0      0        0 eth0
0.0.0.0         10.0.0.31   0.0.0.0         UG    0      0        0 eth0
```

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Conf. of **r1** I

- Let's configure `eth0` in **r1** and try some `pings` from **alice**:

```
r1:~# ifconfig eth0 10.0.0.31/25

alice:~# arp -n
alice:~# ping -c 1 10.0.0.31
PING 10.0.0.31 (10.0.0.31) 56(84) bytes of data.
64 bytes from 10.0.0.31: icmp_seq=1 ttl=64 time=20.6 ms

--- 10.0.0.31 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 20.617/20.617/20.617/0.000 ms
alice:~# arp -n
Address     HWtype  HWaddress           Flags Mask  Iface
10.0.0.31   ether   FE:FD:00:00:02:00   C           eth0
```

- As you can observe, the ping to .31 works.
- If the delete the ARP entry (or wait) and try a ping to .240:

```
alice:~# arp -d 10.0.0.31
alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
From 10.0.0.31 icmp_seq=1 Destination Net Unreachable

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms
alice:~# arp -n
Address     HWtype  HWaddress           Flags Mask  Iface
10.0.0.31   ether   FE:FD:00:00:02:00   C           eth0
```

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Conf. of **r1** II

- Now, let's configure `eth1` in **r1** and the indirect route to the network 10.0.0.192/26:

```
r1:~# ifconfig eth1 192.168.0.1/30
r1:~# route add -net 10.0.0.192/26 gw 192.168.0.2
r1:~# route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
192.168.0.0     0.0.0.0         255.255.255.252 U     0      0        0 eth1
10.0.0.192      192.168.0.2     255.255.255.192 UG    0      0        0 eth1
10.0.0.0        0.0.0.0         255.255.255.128 U     0      0        0 eth0
```

- Let's try again a `ping` from **alice** to .240:

```
alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
From 10.0.0.31 icmp_seq=1 Destination Host Unreachable

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms
```

- Now the problem is that `eth0` of **r2** is not properly configured.

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Conf. of **r2** I

- Let's configure `eth0` in **r2** and try the `ping` from **alice**:

```
r2:~# ifconfig eth0 192.168.0.2/30

alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, 100% packet loss, time 0ms
```

- Notice that **r2** has no route for the network 10.0.0.192/26 but it cannot send an ICMP message to **alice** because it has not any route for 10.0.0.0/25 either.
- Let's configure in **r2** a route to 10.0.0.0/25:

```
r2:~# route add -net 10.0.0.0/25 gw 192.168.0.1
r2:~# route -n
Kernel IP routing table
Destination   Gateway      Genmask          Flags Metric Ref  Use Iface
192.168.0.0   0.0.0.0      255.255.255.252 U     0      0      0 eth0
10.0.0.0      192.168.0.1  255.255.255.128 UG    0      0      0 eth0
```

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Conf. of **r2** II

- If we try the `ping` again:

```
alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
From 192.168.0.2 icmp_seq=1 Destination Net Unreachable

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms
```

- Let's configure `eth1` of **r2** and try the `ping` again:

```
r2:~# ifconfig eth1 10.0.0.202/26
r2:~# route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
192.168.0.0     0.0.0.0         255.255.255.252 U     0      0        0 eth0
10.0.0.192      0.0.0.0         255.255.255.192 U     0      0        0 eth1
10.0.0.0        192.168.0.1     255.255.255.128 UG    0      0        0 eth0

alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
From 192.168.0.2 icmp_seq=1 Destination Host Unreachable

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms
```

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Conf. of **bob** I

- Let's configure `eth0` of **bob** and try the `ping` from **alice**:

```
bob:~# ifconfig eth0 10.0.0.240/26

alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, 100% packet loss, time 0ms
```

- The ICMP message is received by **bob** but he cannot answer to **alice**.
- We have to define a route to 10.0.0.0/25:

```
bob:~# route add -net 10.0.0.0/25 gw 10.0.0.202
bob:~# route -n
Kernel IP routing table
Destination  Gateway      Genmask         Flags Metric Ref    Use Iface
10.0.0.192   0.0.0.0      255.255.255.192 U     0      0        0 eth0
10.0.0.0     10.0.0.202   255.255.255.128 UG    0      0        0 eth0
```

- Finally the `ping` works!!

```
alice:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
64 bytes from 10.0.0.240: icmp_seq=1 ttl=62 time=24.6 ms

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 24.604/24.604/24.604/0.000 ms
```

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Conf. of **bob** II

- But a `ping` from **r1** to **bob** does not work.

```
r1:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 0 received, 100% packet loss, time 0ms
```

- We can add a route to 192.168.0.0/30 in **bob** to fix this:

```
bob:~# route add -net 192.168.0.0/30 gw 10.0.0.202
bob:~# route -n
Kernel IP routing table
Destination  Gateway    Genmask         Flags Metric Ref Use Iface
10.0.0.192   0.0.0.0    255.255.255.192 U     0      0      0 eth0
10.0.0.0     10.0.0.202 255.255.255.128 UG    0      0      0 eth0
192.168.0.0  10.0.0.202 255.255.255.252 UG    0      0      0 eth0

r1:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
64 bytes from 10.0.0.240: icmp_seq=1 ttl=62 time=24.6 ms

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 24.604/24.604/24.604/0.000 ms
```

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Conf. of **bob** III

- A better solution is to add a “default route” (and remove the other indirect routes):

```
bob:~# route add default gw 10.0.0.202
bob:~# route del -net 192.168.0.0/30
bob:~# route del -net 10.0.0.0/25
bob:~# route -n
Kernel IP routing table
Destination  Gateway    Genmask         Flags Metric Ref Use Iface
10.0.0.192   0.0.0.0    255.255.255.192 U     0      0      0 eth0
0.0.0.0      10.0.0.202 0.0.0.0         UG    0      0      0 eth0

r1:~# ping -c 1 10.0.0.240
PING 10.0.0.240 (10.0.0.240) 56(84) bytes of data.
64 bytes from 10.0.0.240: icmp_seq=1 ttl=62 time=24.6 ms

--- 10.0.0.240 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 24.604/24.604/24.604/0.000 ms
```

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Direct Routes I

- Direct routes are not usually configured with `route` because they are automatically created by `ifconfig`.
- However, we can remove and add direct routes also with the `route` command.
- Let's perform a test about direct routes.
- In particular, we want to show you that the routing table is only necessary to send packets but not to receive them.
- To do so, we delete the direct route to 10.0.0.0/25 in **r1**.

```
r1:~# route del -net 10.0.0.0/25
r1:~# ifconfig eth0
eth0      Link encap:Ethernet  HWaddr fe:fd:00:00:02:00
          inet addr:10.0.0.31  Bcast:10.0.0.127  Mask:255.255.255.128
          inet6 addr: fe80::fcfd:ff:fe00:200/64 Scope:Link
          UP BROADCAST RUNNING PROMISC MULTICAST  MTU:1500  Metric:1
          RX packets:67 errors:0 dropped:0 overruns:0 frame:0
          TX packets:34 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4200 (4.1 KiB)  TX bytes:2424 (2.3 KiB)
          Interrupt:5
r1:~# route -n
Kernel IP routing table
Destination   Gateway     Genmask         Flags Metric Ref  Use Iface
192.168.0.0   0.0.0.0     255.255.255.252 U     0      0      0 eth1
10.0.0.192    192.168.0.2 255.255.255.192 UG    0      0      0 eth1
```

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Direct Routes II

- As you can observe, **r1**'s `eth0` still has the IP address but we have removed its associated direct route.
- When you try a `ping` from **alice** to **bob** you will observe in `tap2` that the `echo-request` message crosses **r1**.
- However, the `echo-replay` message cannot be delivered to **alice** and **r1** generates a `destination-unreachable` message.
- Finally, to reconfigure the direct route you can use `ifconfig` with up/down:

```
r1:~# route -n
Kernel IP routing table
Destination   Gateway     Genmask         Flags Metric Ref  Use Iface
192.168.0.0   0.0.0.0     255.255.255.252 U     0      0      0 eth1
10.0.0.192    192.168.0.2 255.255.255.192 UG    0      0      0 eth1
r1:~# ifconfig eth0 down
r1:~# ifconfig eth0 up
r1:~# route -n
Kernel IP routing table
Destination   Gateway     Genmask         Flags Metric Ref  Use Iface
192.168.0.0   0.0.0.0     255.255.255.252 U     0      0      0 eth1
10.0.0.192    192.168.0.2 255.255.255.192 UG    0      0      0 eth1
10.0.0.0      0.0.0.0     255.255.255.128 U     0      0      0 eth0
```

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Direct Routes III

- However, when we set an interface down all the indirect routes that use this interface are also removed from the routing table.
- If you do not desire this behavior, you can use the `route` command and the `dev` option to create direct routes:

```
r1:~# route -n
Kernel IP routing table
Destination   Gateway       Genmask         Flags Metric Ref  Use Iface
192.168.0.0   0.0.0.0     255.255.255.252 U     0      0      0 eth1
10.0.0.192    192.168.0.2 255.255.255.192 UG    0      0      0 eth1
10.0.0.0      0.0.0.0     255.255.255.128 U     0      0      0 eth0
r1:~# route del -net 192.168.0.0/30
r1:~# route -n
Kernel IP routing table
Destination   Gateway     Genmask         Flags Metric Ref  Use Iface
10.0.0.192    192.168.0.2 255.255.255.192 UG    0      0      0 eth1
10.0.0.0      0.0.0.0     255.255.255.128 U     0      0      0 eth0
r1:~# ifconfig eth1 down
r1:~# ifconfig eth1
eth1      Link encap:Ethernet  HWaddr fe:fd:00:00:02:01
          inet addr:192.168.0.1  Bcast:192.168.0.3  Mask:255.255.255.252
          BROADCAST PROMISC MULTICAST  MTU:1500  Metric:1
          RX packets:47 errors:0 dropped:0 overruns:0 frame:0
          TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3460 (3.3 KiB)  TX bytes:2200 (2.1 KiB)
          Interrupt:5
```

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Direct Routes IV

```
r1:~# route -n
Kernel IP routing table
Destination  Gateway Genmask         Flags Metric Ref Use Iface
10.0.0.0     0.0.0.0 255.255.255.128 U     0      0     0 eth0
r1:~# ifconfig eth1 up
r1:~# route add -net 10.0.0.192/26 gw 192.168.0.2
r1:~# route -n
Kernel IP routing table
Destination   Gateway      Genmask         Flags Metric Ref  Use Iface
192.168.0.0   0.0.0.0      255.255.255.252 U     0      0      0 eth1
10.0.0.192    192.168.0.2  255.255.255.192 UG    0      0      0 eth1
10.0.0.0      0.0.0.0      255.255.255.128 U     0      0      0 eth0
r1:~# route del -net 192.168.0.0/30
r1:~# route -n
Kernel IP routing table
Destination   Gateway      Genmask         Flags Metric Ref  Use Iface
10.0.0.192    192.168.0.2  255.255.255.192 UG    0      0      0 eth1
10.0.0.0      0.0.0.0      255.255.255.128 U     0      0      0 eth0
r1:~# route add -net 192.168.0.0/30 dev eth1
r1:~# route -n
Kernel IP routing table
Destination   Gateway      Genmask         Flags Metric Ref  Use Iface
192.168.0.0   0.0.0.0      255.255.255.252 U     0      0      0 eth1
10.0.0.192    192.168.0.2  255.255.255.192 UG    0      0      0 eth1
10.0.0.0      0.0.0.0      255.255.255.128 U     0      0      0 eth0
```

- Finish the configuration of the scenario using the less number of entries possible in the routing tables (use longest match) and making the default routes point to Internet.

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Getting Started I

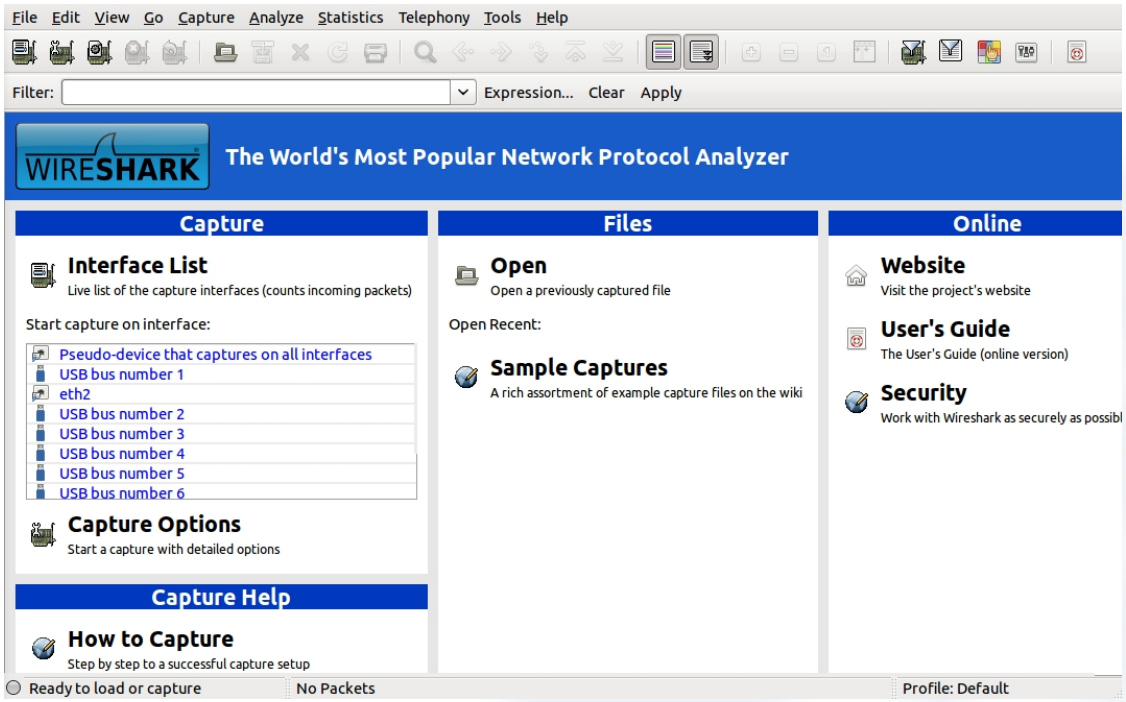
- Wireshark has become the “defacto”, open-source tool for protocol analysis.
- It provides low-level packet filtering, analytical capability and it can be used to store captured traffic in a file for later analysis.
- Once installed, you can run the protocol analyzer from a terminal typing `wireshark`.
- However, if you run Wireshark with an unprivileged user you can only open pcap files (not capture in real time).
- To run wireshark as root in Linux, type (or login as root):

```
$ sudo wireshark
```

- In the initial Wireshark screen you have available the list of all the interfaces of the system.

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Getting Started II



- Click on one of the network interfaces to start capturing packets.

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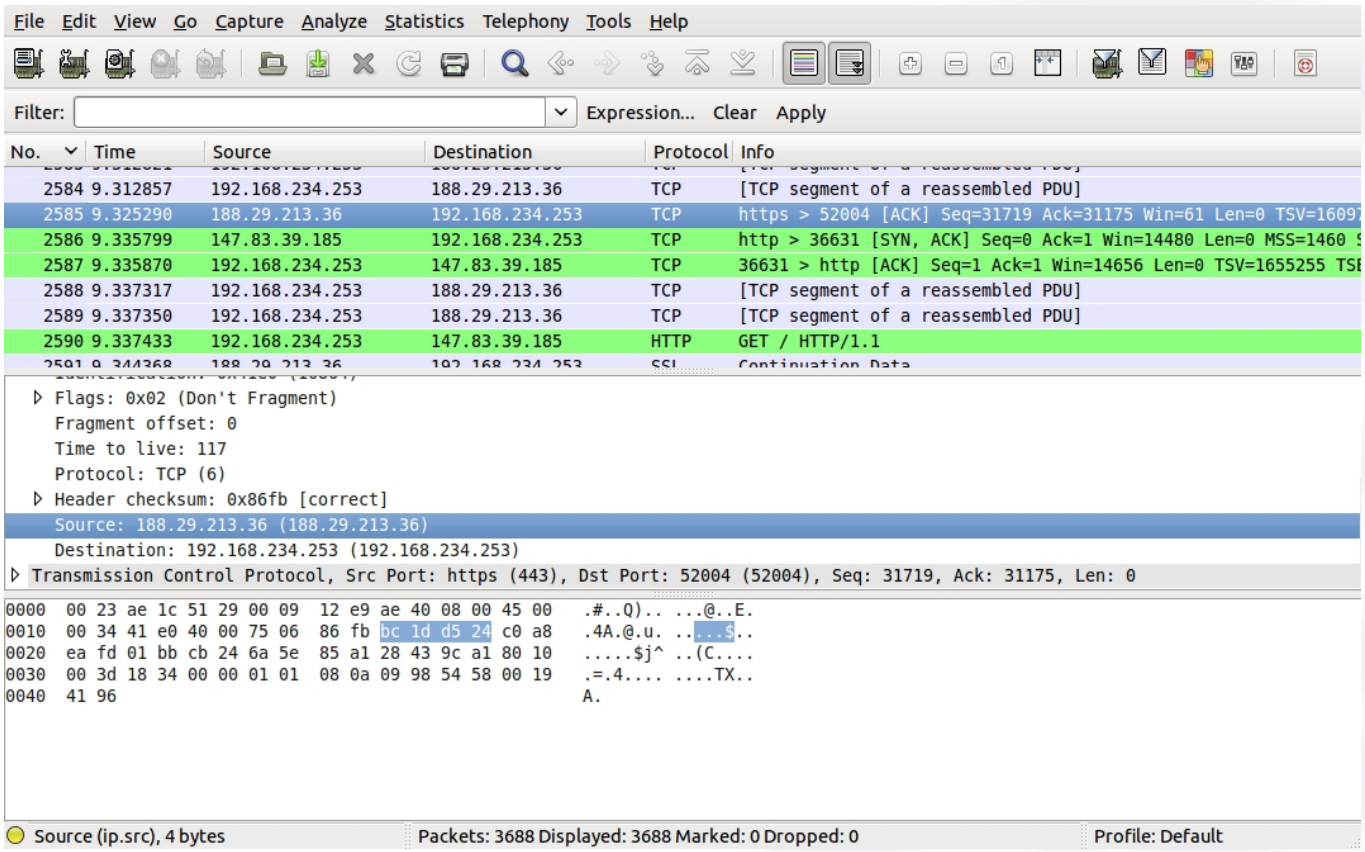
Capturing Packets I

- To capture packets you have to select a network interface.
- When you select an interface, you will be able to see packets moving through that interface.
- When there are captured packets available, you can select a packet and view its fields as decoded by wireshark and also in ASCII and hexadecimal.
- Furthermore, in the bottom left corner of the Wireshark window, the protocol analyzer displays the size of the packet or field selected.

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Capturing Packets II

- In the example of the figure we have selected the IP source address of a packet that as you see occupies 4 bytes as expected.



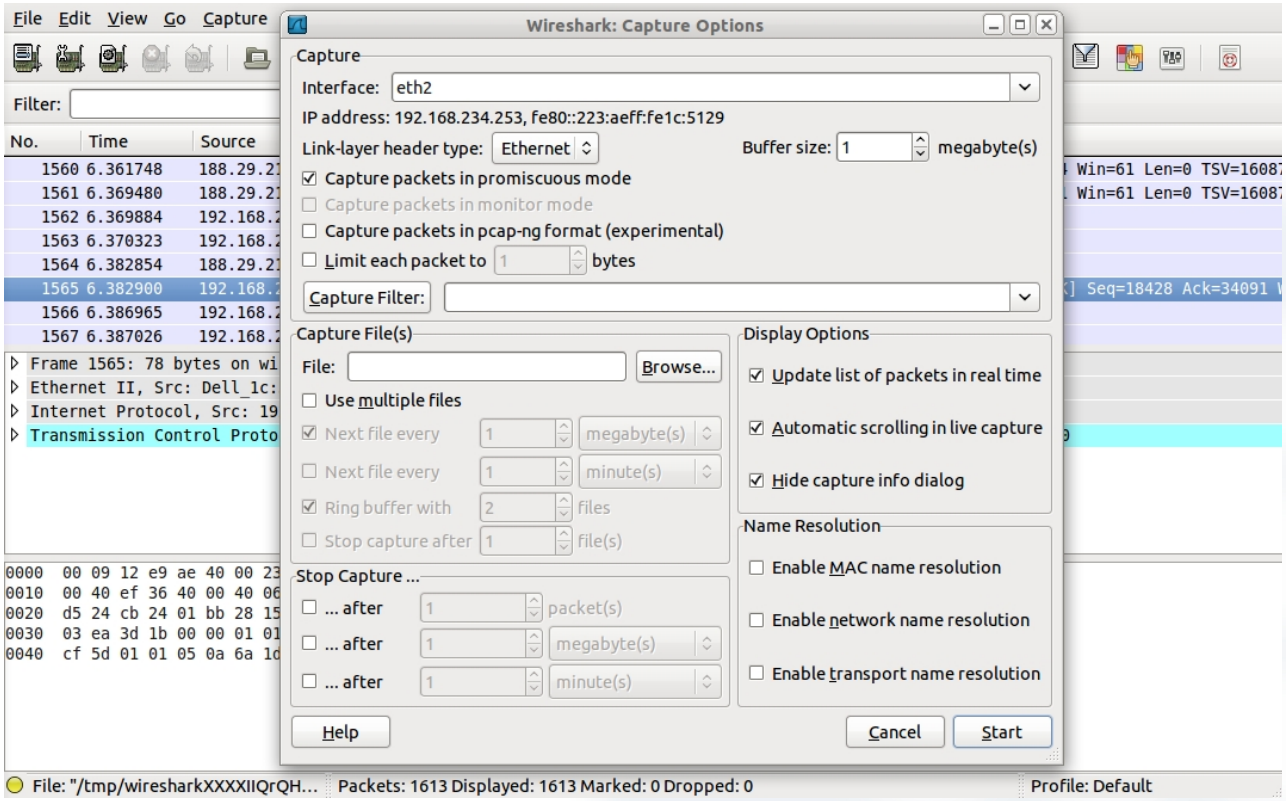
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Capture Options I

- You can also select the capture options before staring the capture.
- In our case, we will unset all the “Name resolution” options and we will set always the ”Capture packets in Promiscuous mode“.
- This mode allows wireshark to capture data link layer frames that are not destined to our host.
- That is to say, frames that have a destination link address that is not the one that has our network interface.

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Capture Options II



- Finally, notice that you can select a ”Capture Filter”.

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Capture filters I

- Capture filters are used to select the data to record in the logs.
- They are defined before starting the capture.
- The basic syntax for creating capture filters is the following:

```
(Parameter Value) Logical_Operation (Parameter Value) ...
```

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Examples I

- To capture only traffic to or from IP address 172.18.5.4, you can type the following capture filter:

```
host 172.18.5.4
```

- To capture traffic to or from a range of IP addresses, you can type the following capture filter (both are equivalent):

```
net 192.168.0.0/24
net 192.168.0.0 mask 255.255.255.0
```

- To capture traffic from a range of IP addresses, you can type the following capture filter (both are equivalent):

```
src net 192.168.0.0/24
src net 192.168.0.0 mask 255.255.255.0
```

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Examples II

- To capture traffic to a range of IP addresses, you can type the following capture filter (both are equivalent):

```
dst net 192.168.0.0/24
dst net 192.168.0.0 mask 255.255.255.0
```

- To capture only HTTP (port 80) traffic, you can type the following capture filter:

```
port 80
```

- To capture non-HTTP and non-SSH traffic on 192.168.0.1, you can type the following capture filter (both are equivalent):

```
host 192.168.0.1 and not (port 80 or port 22)
host 192.168.0.1 and not port 80 and not port 22
```

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Examples III

- To capture all traffic except ICMP and HTTP traffic, you can type the following capture filter:

```
port not 80 and not icmp
```

- To capture traffic within a range of ports, for example TCP ports between 2001 and 2500, you can type the following capture filter:

```
tcp portrange 2001-2500
```

- To capture packets with source IP address 10.4.1.12 or source network 10.6.0.0/16 and having destination TCP port range from 2001 to 2500 and destination IP network 10.0.0.0/8, you can type the following capture filter:

```
(src host 10.4.1.12 or src net 10.6.0.0/16) and
tcp dst portrange 2001-2500 and dst net 10.0.0.0/8
```

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Display Filter I

- Display filters are used to search inside the captured logs.
- They can be applied and modified while data is being captured.
- You may wonder if you should use a capture or a display filter.
- Notice that the goals of the two filters are different.
- The capture filter is used as a first large filter to limit the size of captured data to avoid generating a log too big.

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Display Filter II

- The display filter is much more powerful (and complex); it will permit you to search exactly the data you want.
- Wireshark uses display filters for general packet filtering while viewing and for its coloring rules.
- The basics and the syntax of the display filters are described in the Wireshark User's Guide and you can also use the Analyze menu (option Display filters) to build your display filter.

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Examples I

- To display only HTTP (port 80) and ICMP traffic, you can type the following display filter:

```
tcp.port eq 80 or icmp
```

- To display only traffic between workstations in the LAN 192.168.0.0/16, you can type the following display filter:

```
ip.src==192.168.0.0/16 and ip.dst==192.168.0.0/16
```

- To match HTTP requests where the last characters in the URL/URI are the characters “html”, you can type the following display filter:

```
http.request.uri matches "html$"
```

- Note: The $ character is a regular expression that matches the end of a string, in this case the end of http.request.uri field.

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Streams I

- Another very useful functionality of Wireshark that we are going to use is the “Follow stream”.
- This feature can be selected from the Analyze menu and it works as follows:
  - If you have a TCP or UDP packet selected and you select “Follow TCP stream” or “Follow UDP stream”, it will appear in a separate window all the contents of the data stream to which that packet belongs.
  - In addition, the main display of Wireshark will leave the list of packets in a filtered state, with only those packets that are part of that TCP or UDP stream being displayed.
  - You can revert to your old view by pressing ENTER in the display filter text box, thereby invoking your old display filter (or resetting it back to no display filter).

Streams II

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